

Summary

Applied Scientist | 14+ years in NLP, VLMs, LLMs & Document AI. Skilled in building AI solutions, integrating multi-modal data, and driving business impact through R&D. Thrives in fast-paced, collaborative environments, embracing rapid experimentation to accelerate innovation. Proven leadership in mentoring and contributions to patents/publications. Focused on ensuring trustworthy model robustness, delivering reliable and ethical AI systems.

Awards & Recognitions

- [Social Impact Award](#) - Accesseval a benchmark evaluating disability biases in LLMs (EMNLP 2025).
- Awarded for innovation in AI/ML through building a hybrid recommendation system delivering exceptional CTR and engagement, surpassing Google AutoML performance (Vuclip 2019).

Work Experience

Current Employer: Oracle Cloud Infrastructure

(May 21- Sep 25)

Designation: Senior Applied Scientist

Key Responsibilities:

- [Claim Processing Engine](#) - Build a scalable claim processing engine leveraging LangGraph.
- [Responsible AI Programs](#) - Accessibility ML practices & engineered prompt-injection mitigations guardrail system.
- [LLM & VLM Research](#) - Leading Generative AI R&D, developing LLM and VLM models with PyTorch & Hugging Face to address enterprise AI challenges, optimizing scalability and efficiency.
- [Multi-Modal & Graph-Based AI for Document Intelligence](#) - Developing AI-driven approaches for document understanding by integrating multi-modal deep learning with Transformer and graph neural networks (GNNs).
- [Data Preprocessing & Synthetic Data Generation](#) - Spearheaded data curation, augmentation, and multi-modal dataset creation for Vision & NLP tasks, improving model generalization.
- [Model Trustworthiness & AI Ethics](#) - Established methodologies for AI fairness, robustness, and transparency.
- [Mentorship & Technical Leadership](#) - Guided and onboarded researchers in LLM architectures, fostering innovation and driving best practices in AI model development.

Key Projects & Achievements:

- [Face Recognition and Anti-Spoofing](#) - Achieved close to SOTA performance with face recognition and anti-spoofing. Lead internal data collection and annotation efforts that saved \$400,000 in operational costs.
- [Retrieval-Augmented Generation \(RAG\) Framework](#) - Built an intelligent routing mechanism to optimize QA system efficiency, reducing latency.
- [Generative AI for Code Generation](#) - Developed an LLM for SuiteScript code generation using StarCoderBase, leveraging parameter-efficient fine-tuning to enhance adaptability.
- [Document Understanding & Key-Value Extraction](#) - Developed Oracle Cloud's CustomKV extraction, DL extraction achieving industry parity with Azure and surpassing Google Document AI Custom Model.
- [Data Generation Framework](#) - Designed a multi-modal data-generation pipeline, enabling multi-task learning for Vision & NLP models, boosting overall performance.

Recent Employer: TVS Motor Company

(Oct 20- Apr 2021)

Designation: Principal Data Scientist

- Developed and implemented a [transformer-based recommendation](#) module capable of detecting duplicate queries with 92% accuracy, leading to a 20% reduction in service team workload.

Past Employer: Vuclip (VIU)

(June 18- Sept 20)

Designation: Principal Engineer

- [Hybrid Recommendation System](#): Engineered a hybrid recommendation model combining collaborative and content-based filtering, achieving best-in-class CTR and engagement metrics.
- [Subscription Fraud Detection](#): Reduced fraud leveraging local and global user behavior data.

Past Employer: NASDAQ India, on behalf of Tech Mahindra

(June 17- May 18)

Designation: Technical Lead, Data Analytics

- [Sentiment Analysis for Market Data](#): Built stock market sentiment models for real-time bullish/bearish predictions.

Past Employer: Philips India (Includes two years of Integra Micro System's deputation)

(Jan 10- May 17)

Designation: Data Specialist, Senior Software Engineer, Software Engineer

- Sentiment Analysis: Built classification models for app store reviews, driving actionable customer feedback

Patents & Research Contribution

Accepted Conference Papers:

- [A Diagnostic Framework](#) for Auditing Reference-Free Vision-Language Metrics (AAACL-IJCNLP 2025, Mumbai)
- [AccessEval](#): Benchmarking Disability Bias in Large Language Models (EMNLP 2025, Suzhou)
- [PCRI](#): Measuring Context Robustness in Multimodal Models for Enterprise Applications (EMNLP 2025, Suzhou)
- [RCI](#): A Score for Evaluating Global and Local Reasoning in Multimodal Benchmarks (EMNLP 2025, Suzhou)
- [FlexDoc](#): Parameterized Sampling for Diverse Multilingual Synthetic Documents for Training Document Understanding Models (EMNLP 2025, Suzhou)
- [Who's Asking?](#) Investigating Bias Through Lens of Disability-Framed Queries (ICCV 2025, Hawaii)
- [MVTamperBench](#): Evaluating Robustness of Vision-Language Models (ACL 2025, Vienna)
- [Hard Negative Mining](#) for Domain-Specific Retrieval in Enterprise Systems (ACL 2025, Vienna)
- [SweEval](#): Do LLMs Really Swear? A safety benchmark testing LLM limits (NAACL 2025, New Mexico)
- [Multi-Turn Online Conversations](#) Novel Dynamic Routing and Feedback Adaptation (NAACL 2025, New Mexico)
- [FS-DAG](#): Few-Shot Domain Adapting Graph Network (COLING 2025, Abu Dhabi)
- [A Bipartite Graph Approach](#) to linking selection marks and text (ICDMAI 2025, Kolkata)
- [IJAIML 04_01_003](#): Review of reference generation methods in large language models

Preprints:

- [2510.04230](#): Pushing On Multilingual Reasoning Modeity in Large Language Models.
- [2508.15830](#): DAIQ: Auditing Demographic Attribute Infels With Language-Mixed Chain-Of-Thought
- [2511.10519](#): Say It Differently: Linguistic Styles as Jailbreak Vectors
- [2509.00963](#): Who Gets Left Behind? Auditing Disability Inclusivrence from Question in LLMs.
- [2502.13108](#): Clinical qa 2.0: Multi-task learning for answer extraction and categorization.
- [2502.13108](#): Improving Clinical Question Answering with Multi-Task Learning.
- [2412.17759](#): Survey of Large Multimodal Model Datasets, Application Categories and Taxonomy.
- [2412.03590](#): Enhancing Document AI Data Generation Through Graph-Based Synthetic Layouts.

Patents:

- [US20250238614A1](#): Dynamic vocabularies for conditioning a language model.
- [US20250209085A1](#): Table Extraction from Images Using Language Models.
- [US20250078556A1](#): Techniques of information extraction for selection marks.
- [US20250078555A1](#): Techniques of information extraction for selection marks.
- [US20250014374A1](#): Out of distribution element detection for information extraction.
- [US20240005640A1](#): Synthetic document generation pipeline for training artificial intelligence models.
- [US20240289551A1](#): Domain adapting graph networks for visually rich documents.
- [US20200252802A1](#): Apparatus and method for fraud detection.
- ORC26140658-US-NPR: Safety-Aware Adaptive Few-Shot Prompting
- ORC25140081-US-NPR: RES - Robustness Evaluation System
- ORC25139913-US-PSP: Hybrid AI For Responsive Multi-turn Online Conversations with novel Dynamic Routing and Feedback Adaptation.
- ORC25139817-US-PSP: Evaluating MLLMs Performance against Image Context.
- ORC25139275-US-NPR: A system and approach for improved language model performance with Dynamic Feature Selection across Transformer layers.
- ORC25139514-US-NPR: Method and apparatus for Multi block image processing.
- ORC24138454-US-NPR: Taxonomic Negative Sampling-Based Machine Learning.
- ORC24138489-US-NPR: Context-Driven Fine-Tuning for Reliable Retrieval-Augmented Generation.

Skill Set

- **Programming language and Machine Learning Framework**: Python, PyTorch, TensorFlow
- **Deep learning**: Feed Forward Neural Network, Convolutional Neural Network, Recurrent Neural Network, Transformer
- **Cloud Computing Service**: OCI, GCP Big query and Document AI, Azure Form Recognition

Academic Background

- M-Tech Program in Software System with Data Analytics Specialization from BITS, Pilani (2016-2018)
- B-Tech from Electronics & Communication Engineering from BPUT, Odisha (2003-2007)